A new contrast enhancement method for Color dark and low-light images

1st Meysam Alavi [1], 2nd Mehrdad Kargari [2]

[1] Information Technology dept.
Tarbiat Modares
Tehran, Iran
meysamalavi@modares.ac.ir

[2] Information Technology dept.
Tarbiat Modares
Tehran, Iran
M_kargari@modares.ac.ir

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***Abstract*— Image contrast enhancement is a preprocessing phase that improves the performance of image processing applications such as pattern recognition and computer vision. Many images have poor quality due to low- luminance and low-contrast, which must be changed before further processing. It is significant for medical imaging because the low-light intensity makes it challenging to diagnose and analyze specific diseases accurately. In addition, when the depth information of a low-light image is unknown, the drawback of illumination enhancement becomes very challenging. Since Contrast enhancement of low-light images with non-uniform illumination is complex, it may lead to inefficient contrast enhancement methods. In other words, in such cases, if contrast enhancement methods are used to increase the contrast of dark areas, the bright regions become over-enhanced, which may lead to the disappearance of the details of these areas. Given this problem, in this paper, a new method is proposed to increase the image contrast that can improve the contrast in both areas of the image without adversely affecting the details of the image. The result represented the efficiency of proposed methods compared to other image enhancement methods based on full-reference and no-reference metrics such as PSNR, MSE, NIQE, and BRISQUE.**

Keywords— image processing, image enhancement, contrast enhancement, low-light images.

I. INTRODUCTION

Image enhancement plays a substantial role in image processing and analysis. Improving the image includes increasing contrast, increasing the edge, reducing noise, which enhances the perception of the human supervisor of the information in the image[1]. Contrast enhancement is considered a commonly used image enhancement method. Contrast is defined as segregation in intensity level between an image's bright and dark regions[2]. Problems like device malfunction, the inexperience of taking photographs, and imaging with low or high illuminance intensity, may lead to reduced contrast. As a result, degraded images may disturb the visual aesthetics[3].

Insufficient contrast in images reduces the ability of the observer to understand and analyze the image[4]. It is significant for medical imaging because the low-light intensity makes it challenging to diagnose and analyze specific diseases accurately[1]. Furthermore, in dark images with non-uniform illumination, low contrast leads to the disappearance of the details in both the dark region and the bright region. In such cases, contrast enhancement methods may lead to over-enhanced in the bright area. Hence some details of the bright region are deleted[5]. Dark or weak illumination image enhancement is one of the challenging research areas which have applications in night vision, forensic applications, and underwater enhancement[6]. Owing to the significance of this issue, different methods have been proposed by researchers. Although these methods have tried to improve the contrast, some cases have problems such as loss of image detail, lack of preservation of structural features of the image, and increased noise. Given the importance of the issue, this paper a method for improving contrast based on the separation of red, green, and blue channels of an image and improving each channel, will present that while improving image contrast and noise reduction, the structural features of the image are preserved to an acceptable level. The results demonstrate the efficiency of the proposed method compared to other methods.

The organization of this paper is as follows: In Section 2, an overview of some techniques of image contrast enhancement is addressed. The proposed method describes in Section 3. Section 4 presents experimental results. Finally, in Section 5, the conclusion and future work are drawn.

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II. LITERATURE REVIEW

In recent years, researchers have proposed different methods of contrast enhancement, which provide better images for various applications of image processing. Generally, these methods are categorized into global, local, and hybrid contrast enhancement[7]. On the other hand, these methods are categorized into direct and indirect methods [8]. Most of the methods used in the previous research belong to the second group. One of the most favorite techniques is histogram equalization(HE), which can be global or local. The aim of HE is the intensity values distribution in an image such that the lower contrast regions can achieve higher contrast [9]. Paleness in some areas, loss of some details, and increased noise in some image areas are the disadvantages of this method [10]. Contrast Limited Adaptive Histogram Equalization (CLAHE) to overcome the problems of the HE is presented, which is a local HE method[11]. CLAHE provides maximum entropy while flattening the histogram; therefore, it is more suitable for medical images, especially for X-Ray images, where the low contrast is a challenge. [9]. In the following years, based on the Histogram equalization, various methods were proposed, which can be referred to as BBHE[12], DSIHE[13], RMSHE[14], RSIHE[15], MMBEBHE[16], BPDHE[17]. In[18], Chiu et al. proposed an enhancement method based on the adaptive gamma correction method for low-light images, which used weight distribution to correct the pixel intensity of the input image. This method faced over-enhancement in bright regions of the image. Also, in [19], a technique based on gamma correction to increase the brightness level of a low-contrast image and, Consequently, correct the probability distribution of bright pixels is presented. In [20] a robust Retinex model is presented that adds noise terms to handle low-light image enhancement in the case of drastic noise. Results of the research show this method has good affective in low-light image enhancement. Therefore, it can be generalized to enhance images in underwater, remote sensing, and hazy or dusty conditions. In [21], a fusion-based method to improve low-light images is proposed that can handle various low-light parameters in the same, which images can be enhanced in different weak illumination conditions such as non-uniform illumination, backlighting, and nighttime. Shi et al. proposed an image enhancement method that used a dual-channel method to enhance nighttime low illumination images. The results demonstrate substantial validity of the approach both visually and quantitatively compared to other available methods[22]. Regarding the literature review, this paper presents a new method to enhance the image, which we are able to enhance the contrast in both regions of the image without changing the main points. It also preserves brightness and structural image features while noise is reduced.

III. THE PROPOSED METHOD

In this section, the proposed method is presented. In this method, First, the original image in the RGB-color space is divided into three red, green, and blue channels. Then the algorithm is applied to each RGB channel separately. For each RGB channel of the input image, fuzzification is done using the formula:

$$\mu(g_{ij}) = \frac{g - g_{\min}}{g_{\max} - g_{\min}} \tag{1}$$

Where g is the intensity of the image ranges from 0 to L-1. Minimum and maximum values of the intensity for each RGB channel are illustrated as $g_{\min}$ and $g_{\max}$. The new membership function has been demonstrated with μ_{new}, which is composed of two parts: natural fuzzy complement and Sugeno. These a mentioned in Equations (2) and (3), respectively [23].

$$\bar{\mu} = 1 - \mu \tag{2}$$

$$C_{\lambda}(\mu) = \frac{1-\mu}{1+\lambda\mu} \tag{3}$$

μ_{new}, Based on Equations (2), (3), and (4), Eq. (5) can be concluded.

$$\mu_{new} = 1 - C_{\lambda}(\mu) \tag{4}$$

$$\mu_{new} = \frac{\mu(1+\lambda_{Opt})}{1+\lambda_{Opt}\mu} \tag{5}$$

In the proposed method, an index composed of three metrics, Feature similarity Index (FSIM), structural similarity index (SSIM), and Absolute mean brightness error (AMBE), is defined to improve the contrast of an image while preserving brightness and structural image features. This index is shown in Eq. (6).

$$\text{Evaluation Index} = \frac{FSIM * SSIM}{AMBE} \tag{6}$$

Using SSIM, the structure of the input image with the enhanced image is compared[24], calculated by Eq. (7).

$$SSIM(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{7}$$

Where μ_x, μ_y are the mean intensity of the original and enhanced image, respectively, and σ_x^2, σ_y^2 and σ_{xy} are the variances and the covariance, respectively. A higher value of SSIM shows better performance[25]. The Maximum SSIM value is 1. The FSIM is used as a qualitative metric for the enhanced images. It is calculated as follows:

$$FSIM = \frac{\sum S_L * PC_m}{\sum PC_m} \tag{8}$$

PC And *S* are the maximum local adaptation and the similarity matrix, respectively[26]. The AMBE is calculated as below:

$$AMBE = \left| M(X) - M(Y) \right| \tag{9}$$

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The low value of AMBE illustrates that the average brightness of the enhanced image is close to or equal to the average brightness of the input image[27]. Changing λ leads to creating images with different brightness levels. In other words, changing λ from -1 to infinity creates completely dark to completely white images, and consequently, one of the values in the range of -1 to infinity leads to the optimal state of image contrast. This value of λ is indicated optimal λ and is defined as λ_{opt} . After increasing the λ value, the image is created based on the μ_{new} . Then the evaluation index is calculated and compared with the threshold. If the indicator is higher than the defined threshold, the λ value increases again. This process continues, and when it stops, the last λ value is selected as the optimal λ . Finally, the output image is created based on the optimal λ_{opt} value using Eq. (5). Based on the experimental criterion, the threshold value is considered 0.05, which has the best result. The mentioned process is performed separately for each red, green, and blue channel, and finally, the outputs are merged, and image contrast is enhanced. The pseudo-code of the proposed method is shown in Algorithm 1.

Algorithm 1

Input: Color Image
Output : Enhanced Image
For R,G,B channel of an image
Start
Step1 : the input image is fuzzified using Eq.(1)
Start loop
Step 2 : Increase λ and Calculate μ_{new} using Eq.(5) .
Step 4: Calculate $\frac{FSIM * SSIM}{AMBE}$
if $\frac{FSIM * SSIM}{AMBE} <$ threshold
Go to Step 2
End loop
Step 4 : Determine optimal λ_{opt}
Step 5 : Calculate μ_{new} based on λ_{opt}
Step 6 : Image Defuzzification
End
Step 7: Merge of outputs

A. *Evaluation metrics*

Given the close relationship between image evaluation and human visual perception, finding a universal standard for quantifying the quality of an improved image is difficult[21]. In general, image quality assessment (IQA) is categorized into no reference and full reference. To evaluate the quantitative analysis of the proposed method with other methods are considered, different metrics like Entropy, PSNR, MSE, NIQE[28], and BRISQUE[29].

Entropy is defined as in Eq. (10).

$$Entropy = -\sum_{i=0}^{i-1} p(i)\log_2 p(i) \tag{10}$$

According to entropy theory, the more extensive and detailed the information in an image, the greater the entropy (IE) [30]. The PSNR of an image is used for measuring the image denoising effect[31], as shown in Eq. (11). In which L is 256 for 8-bit gray-scale images and MSE is the mean squares error, as shown in Eq. (12). According to Eq. (11), the higher the PSNR value, the lower the mean square error, so the better the image quality[25].

$$PSNR = 10 * \log_{10}\left(\frac{(L-1)^2}{MSE}\right) \tag{11}$$

$$MSE(X,Y) = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N}\left|X(i,j) - Y(i,j)\right|^2}{M \times N} \tag{12}$$

BRISQUE is considered as the no-reference image quality category metric, which compares input images to a default model computed from images of natural scenes with similar distortions[32]. The smaller score of NIQE and BRISQUE illustrates better perceptual quality [33].

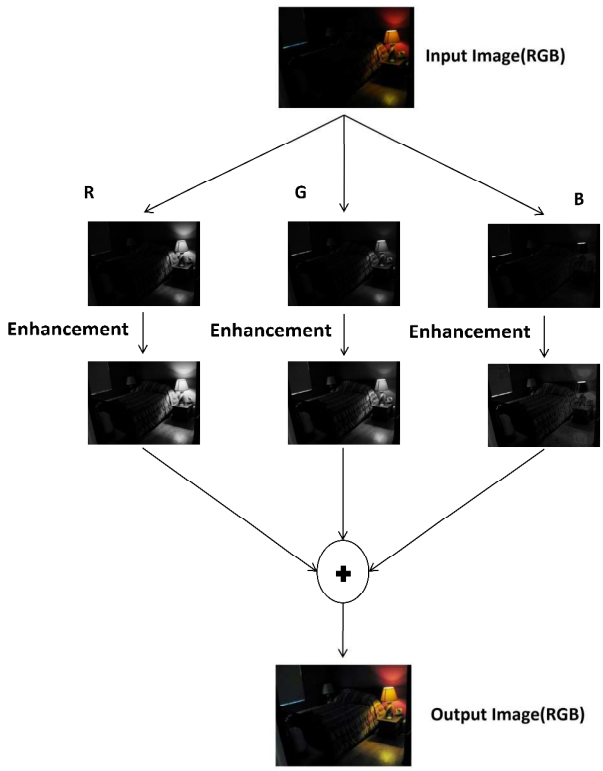

Fig. 1. Outline of the proposed method

IV. EXPERIMENTAL RESULTS

In this research, the Exdark dataset is used for the experimental analysis. The dataset is available on GitHub [34]. To the assessment of the proposed method, we used 45 images, which included dark and low-light images. Also, the proposed method is compared with HE, method of [21], method of[20], AGC[3]. The comparison of average results for images based on evaluation metrics (no reference and full reference) is shown in Figures 2 to 7.

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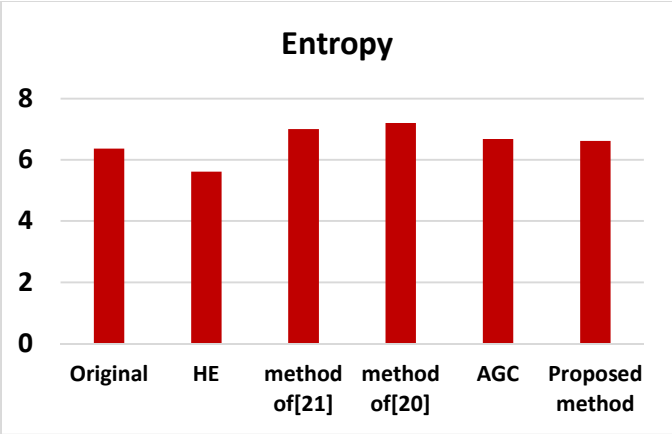

Fig. 2. Entropy average for total images in a different method

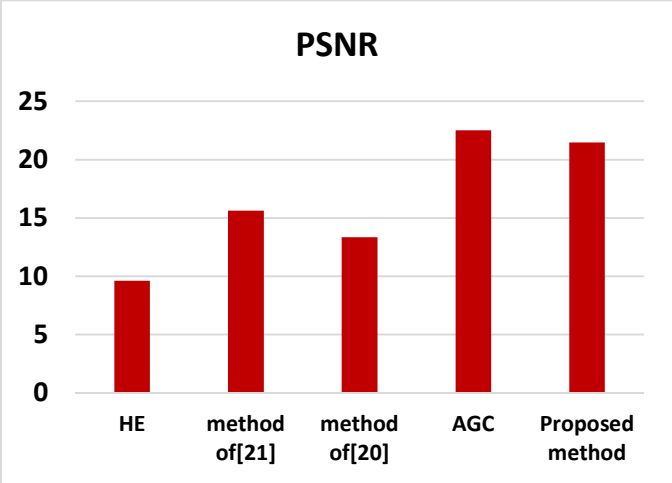

Fig. 3. PSNR average for total images in a different method

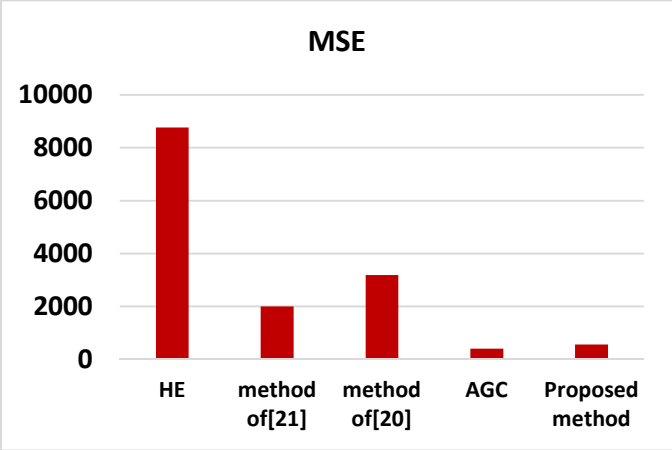

Fig. 4. MSE average for total images in a different method

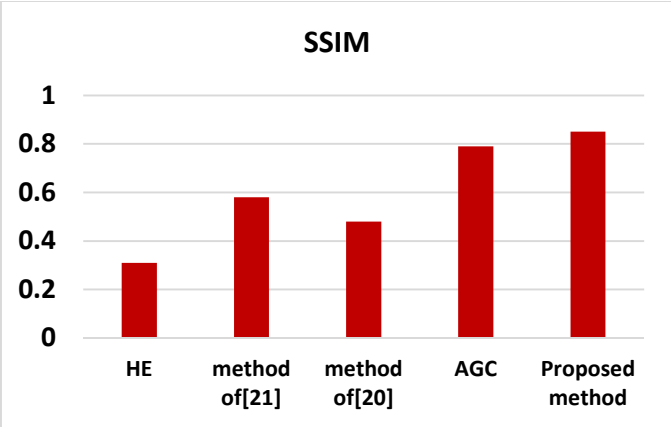

Fig. 5. SSIM average for total images in a different method

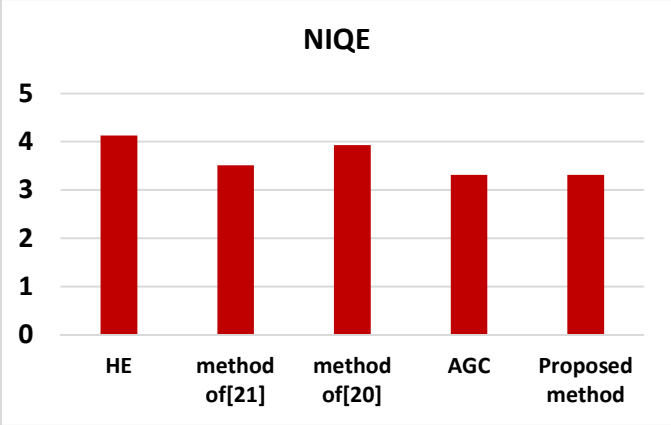

Fig. 6. NIQE average for total images in a different method

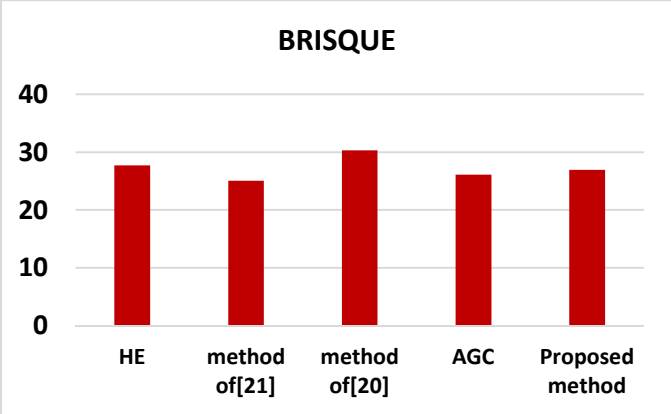

Fig. 7. BRISQUE average for total images in a different method

Experimental results show the efficiency of our method in dark images is better than other methods. In contrast, the AGC method in the low-light image has better performance. Furthermore, as seen in the average case, the proposed method and AGC have similar performance in different metrics. In the SSIM metric, the proposed method has better performance than AGC; that is, the proposed method, better than AGC, preserves the structure of the image. Figure 8 shows some dark images of different enhancement methods compared to the proposed method. In this figure, (A) is the original image, (B) HE, (C) method of[21], (D) method of[20], (E) AGC, and (F) is the proposed method.

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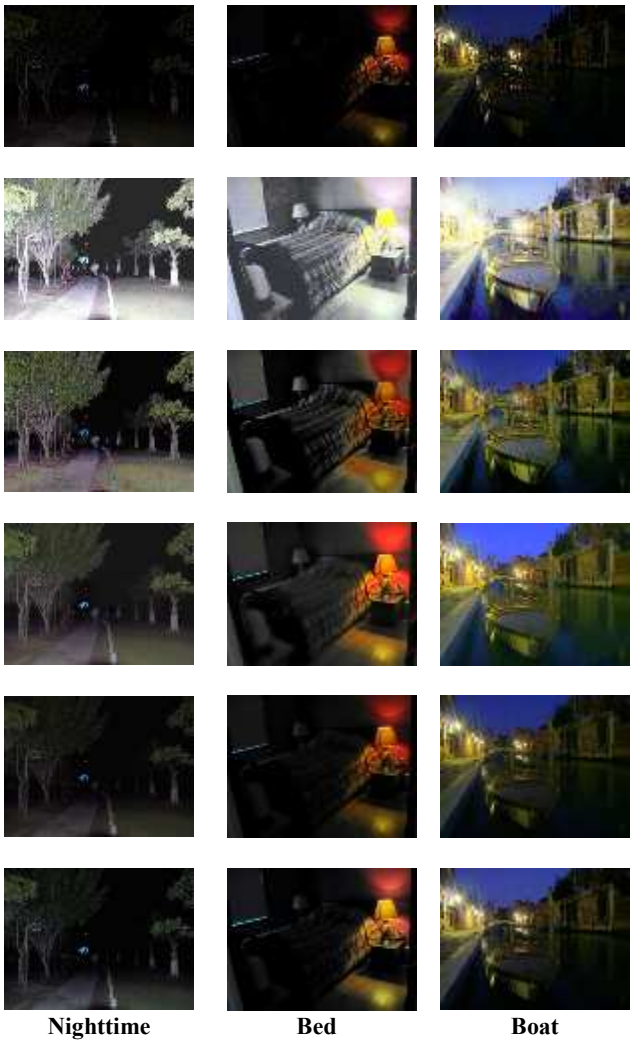
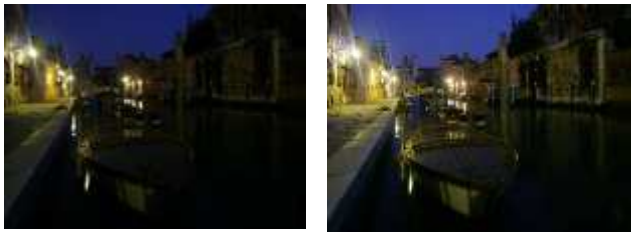


Fig. 8. Results of different enhancement methods

Also, a comparison of the original images and the proposed method can be seen in Fig. 9.



(a) (b)

Fig. 9. Original image (a) vs. Proposed method(b)

Figure 10 shows some low-light images of different enhancement methods compared to the proposed method. In this figure, (A) is the original image, (B) HE, (C) method of[21], (D) method of[20], (E) AGC, and (F) is related to the proposed method.

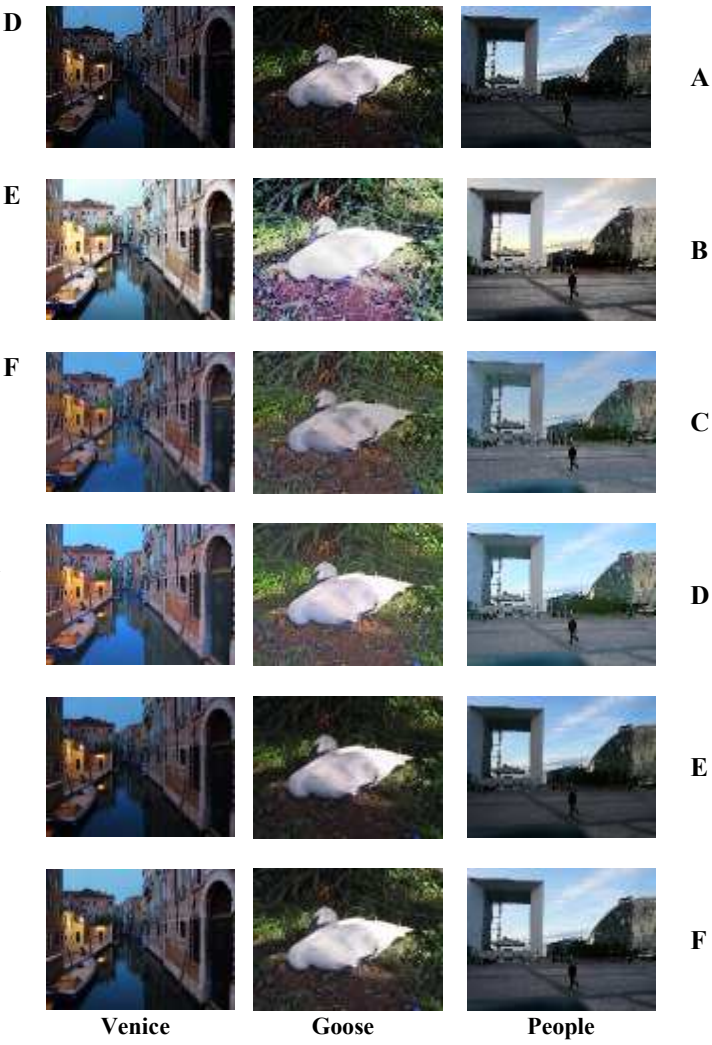


Fig. 10. Results of different enhancement methods

Figure 11 shows a comparison of the original images and the proposed method for low-light images.

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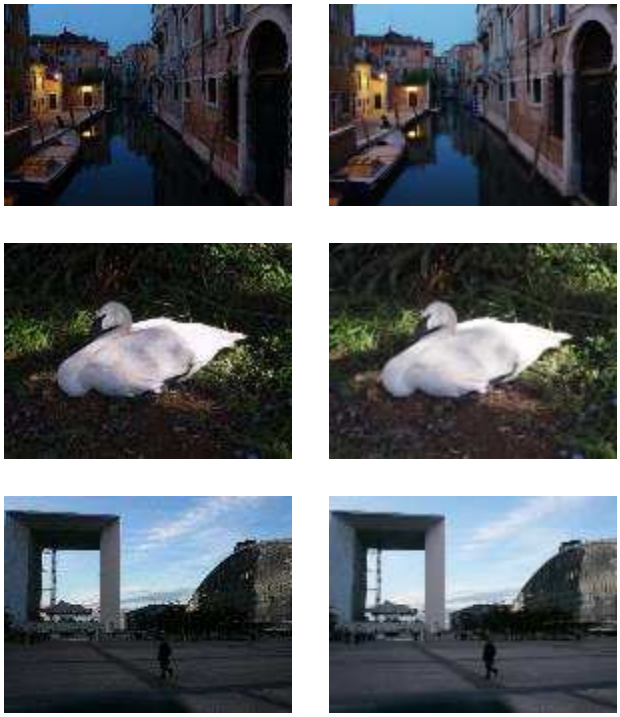

(a) (b)

Fig. 11. Original image (a) vs. Proposed method(b)

V. CONCLUSION AND FUTURE WORK

Contrast enhancement is One of the most challenging issues in image processing. This issue, especially for low-light images with non-uniform illumination, is complex. In this paper, a method for improving contrast dark and low-light images based on the separation of red, green, and blue channels of an image and improving each channel, presented that while improving image contrast and noise reduction, the structural features of the image preserved to an acceptable level. In this research, the defined threshold is a fixed value that can be determined automatically based on the input image. Also, apply of this method for color images contrast enhancement using converted RGB to HSV space can be considered as a continuation of this research.

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